In all results we see that the gapless excitation bands emerge from the Γ point, and that these all have a linear dependence on the excitation wavevector near Γ. For the dipolar case shown in Figs. 1(b) and (c), the lowest branch (close to Γ) is a transverse excitation of the crystal, which has no weight in the dynamic structure factor[4]. The next branch is the second sound or phase mode, which has a weak density contribution. These two lowest branches have similar speeds of sound (i.e. slope near Γ), and as ϵ_{dd} increases the second sound speed decreases and becomes slower than the transverse speed of sound [see subplot (d)]. The third gapless branch of the spectra is known as first sound and is a longitudinal density wave. This excitation branch rises much more steeply than the other two, i.e. exhibiting a much higher sound speed. It is instructive to compare these results to the spectra along the three symmetry directions calculated for the 2D soft-core supersolid (also studied in Ref. [28]). As shown in Fig. 2, the order of the three branches is preserved while varying Λ, with the gapless transverse branch always sandwiched in between the second sound mode and the first sound mode. In Fig. 3, we show three exemplar modes that have the same quasimomentum $\mathbf{q}$ but belong to the three different gapless energy bands. To highlight the character of each mode, we plot the change in density $\Delta|\psi|^2 = \mathcal{N}^{-1}|\psi + c_{\nu\mathbf{q}}u_{\nu\mathbf{q}}(\mathbf{x}) - c^*_{\nu\mathbf{q}}v^*_{\nu\mathbf{q}}(\mathbf{x})|^2 - |\psi|^2$, obtained by subtracting the ground state density from the perturbed density, normalized by $\mathcal{N}$ to the same value as the ground state. Here, we observe that the second sound branch (a) keeps the location of the density maxima fixed, but density changes along the direction of propagation by atom tunnelling between sites. The transverse sound branch (b) causes a shearing of the lattice sites (i.e. transversal motion). The first sound branch (c) exhibits longitudinal displacement of the lattice sites, consistent with a classical crystal excitation. Notice that, to help the visualization, the quasimomentum $\mathbf{q}$ of the selected modes is chosen along along the x-axis, which is equivalent to the $\Gamma - K$ symmetry direction in Fig. 2.

IV. HYDRODYNAMIC DESCRIPTION

The features of the linear part of the excitation spectrum of a 2D supersolid, i.e., the low-energy region for small momenta ($q \ll \pi/a$), are well captured by a hydrodynamic description [14, 42]. This approach is based on long-wavelength perturbations of the ground state, obtained by applying small variations to the three fields associated with conserved quantities and broken symmetries: the change in average density $\delta\rho$, the displacement field $u_{i=\{x,y\}}$ that deforms the planar coordinates as $x_i \to x'_i = x_i + u_i$, and the superfluid phase field ϕ. Before writing the Lagrangian density for a 2D supersolid to obtain the three speeds of sound, it is useful to extract the elastic parameters of the system. Later, these will appear as coefficients of the Lagrangian.

[4] Note that at low q the first and second sound branches have small weight due to the density fluctuations being suppressed, but for the transverse excitation branch the weight vanishes at all q.

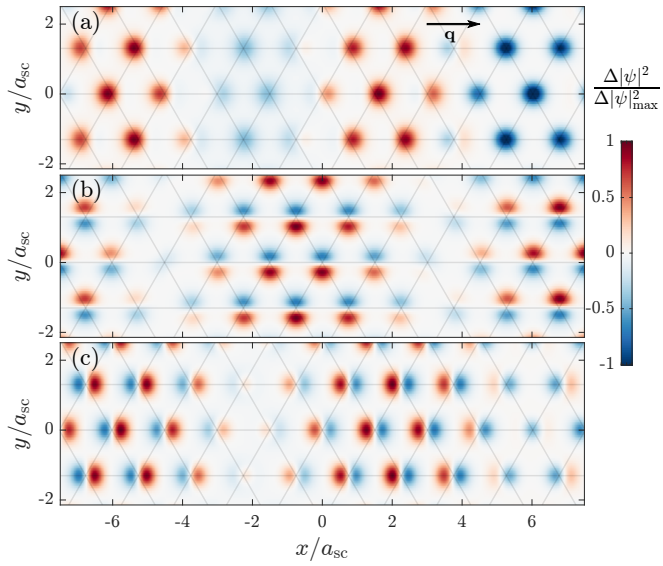

FIG. 3. Effect of excitations from the lowest three gapless bands on the supersolid density profile. The change in density $\Delta|\psi|^2$ (divided by its maximum value) from the addition of an excitation in the (a) second sound, (b) transverse, and (c) first sound bands. The lines intersect at the locations of the density peaks of the ground state. Results for 2D soft-core mode using $\mathbf{q} = (0.4, 0)/a_{sc}$ and $\Lambda = 42.5$. [cf. Fig. 2(c)].

A. Elastic parameters

The first parameter is the superfluid fraction determined by the energetic response of the ground state to changes in the superfluid velocity (see Ref. [36, 43, 44]). In general the superfluid fraction is a tensor given by

$$f_{s,ij} = \frac{1}{m\rho}\frac{\partial^2\mathcal{E}}{\partial v_i \partial v_j}, \tag{4}$$

where the indices $i, j = \{x, y\}$ denote the planar coordinates. For the triangular ground state this tensor is isotropic, i.e. $f_{s,ij} = f_s\delta_{ij}$, and we can simply refer to the superfluid fraction as a scalar value f_s. As a consequence, the average superfluid density $\rho_s = f_s\rho$, and the average normal density $\rho_n = (1 - f_s)\rho$, are also isotropic quantities.

The other elastic parameters involve the dependence of the energy density on the areal density and lattice vectors. The second derivative of $\mathcal{E}$ with respect to the average density ρ defines

$$\alpha_{\rho\rho} = \frac{\partial^2\mathcal{E}}{\partial\rho^2}, \tag{5}$$

which relates to the isothermal compressibility at constant strain:

$$\tilde{\kappa} = \frac{1}{\rho^2\alpha_{\rho\rho}}. \tag{6}$$

The elastic tensor associated with the crystalline structure is

given by

$$C_{ijkl} = \frac{\partial^2 \mathcal{E}}{\partial u_{ij} \partial u_{kl}}, \tag{7}$$

where $u_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ is the strain tensor arising from the displacement field $\mathbf{u}$. We obtain this tensor by evaluating how the ground state energy density changes with the lattice vector deformations $a'_{\sigma=1,2,i} = (\delta_{ij} + u_{ij}) a_{\sigma,j}$ (using the Einstein summation convention). We have verified that the elastic tensor for the triangular ground state is isotropic [45], being of the form

$$C_{ijkl} = \tilde{\lambda}\delta_{ij}\delta_{kl} + \tilde{\mu}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \tag{8}$$

where $\{\tilde{\lambda}, \tilde{\mu}\}$ are the Lamé parameters. We denote $\alpha_{uu} = \tilde{\lambda} + 2\tilde{\mu}$, which is the diagonal element of the elastic tensor (i.e. $C_{xxxx} = C_{yyyy}$), also known as the longitudinal or P-wave modulus. We also have a special interest in the shear modulus $\tilde{\mu}$, which is given by off-diagonal tensor elements, such as C_{xyxy}.
Finally, we consider the density-strain coupling parameter given by the mixed partial derivative

$$\alpha_{\rho u} = \frac{\partial^2 \mathcal{E}}{\partial u_{ii} \partial \rho}, \tag{9}$$

which describes the coupling between changes in average density and the unit cell area.

B. Hydrodynamic theory

We now give the quadratic Lagrangian density for the fields $\delta\rho$, u_i, $\delta\phi$, and solve the Euler-Lagrange equations to extract the speeds of sound. The quadratic Lagrangian density for a supersolid, introduced by Yoo *et al.* [14], reads:

$$\begin{aligned}\mathcal{L}_{\mathrm{SS}}^{\mathrm{quad}} =& -\hbar\delta\rho\partial_t\phi - \rho\frac{\hbar^2}{2m}(\partial_i\phi)^2 - \alpha_{\rho u}\delta\rho\partial_i u_i \\ &- \frac{1}{2}\alpha_{\rho\rho}(\delta\rho)^2 + \frac{1}{2}m\rho_n\left(\partial_t u_i - \frac{\hbar}{m}\partial_i\phi\right)^2 \\ &- \frac{1}{2}C_{ijkl}\partial_i u_j \partial_k u_l.\end{aligned} \tag{10}$$

From the solution of the Euler-Lagrange equations (see Appendix D), one gets the three speeds of sound:

$$mc_1^2 = \frac{1}{2}(a_\Delta + \sqrt{a_\Delta^2 - 4b_\Delta}), \tag{11}$$

$$mc_2^2 = \frac{1}{2}(a_\Delta - \sqrt{a_\Delta^2 - 4b_\Delta}), \tag{12}$$

$$mc_t^2 = \frac{\tilde{\mu}}{\rho_n}, \tag{13}$$

where we have defined

$$a_\Delta = \rho\alpha_{\rho\rho} - 2\alpha_{\rho u} + \frac{\alpha_{uu}}{\rho_n}, \tag{14}$$

$$b_\Delta = \frac{\rho_s}{\rho_n}\left(\alpha_{\rho\rho}\alpha_{uu} - \alpha_{\rho u}^2\right). \tag{15}$$

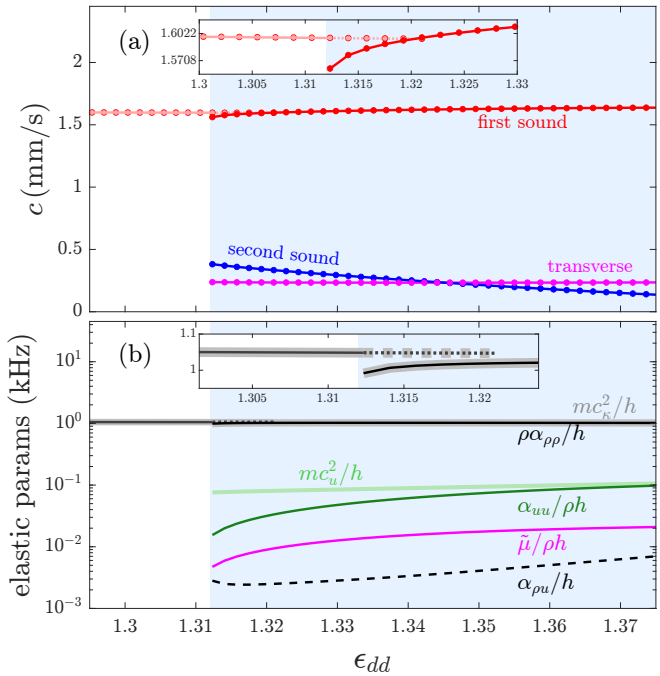

FIG. 4. Elastic properties and speeds of sound for a planar dipolar BEC at the first-order transition to a supersolid. (a) Speeds of sound. We show the speed of first sound in the uniform (light red) and supersolid (red) phases, and the speeds of transverse and second sound in the supersolid phase (as labelled in subplot). Results from the BdG analysis are shown as lines, and the hydrodynamic predictions using the elastic parameters from the ground state calculations are shown as markers. (b) The elastic parameters characterising the dependence of the ground state energy on lattice and density changes. The characteristic energies mc_u^2 (17) (light green) and mc_κ^2 (16) (grey) are also shown for reference. In all subplots the uniform-to-supersolid transition point is indicated by the white to light-blue shaded area in the background. The uniform results are continued below the transition (where it is metastable) until it becomes dynamically unstable when the roton softens (at $a_s \approx 99a_0$). Insets to subplots (a) and (b) show the behaviour near the transition in detail. The other parameters are the same as in Fig. 1.

The quantities c_1, c_2 and c_t represent the first, second and transverse speeds of sound, respectively, and they are fully determined by the elastic coefficients and density of the system.

C. Hydrodynamic results

In Figs. 4 and 5 we consider properties of the dipolar and soft-core systems across the superfluid to supersolid phase transition. The speeds of sound extracted from linear fits to the $q \to 0$ behavior of BdG energies calculated for the gapless branches are shown in Figs. 4(a) and 5(a). The uniform superfluid state only has a single gapless branch: first sound (c_1). In the transition to a supersolid there is almost no change in c_1 for the dipolar case, while a significant upward jump in c_1 occurs for the soft-core case. We also note that in the dipolar system c_1 is higher relative to the other speeds of sound

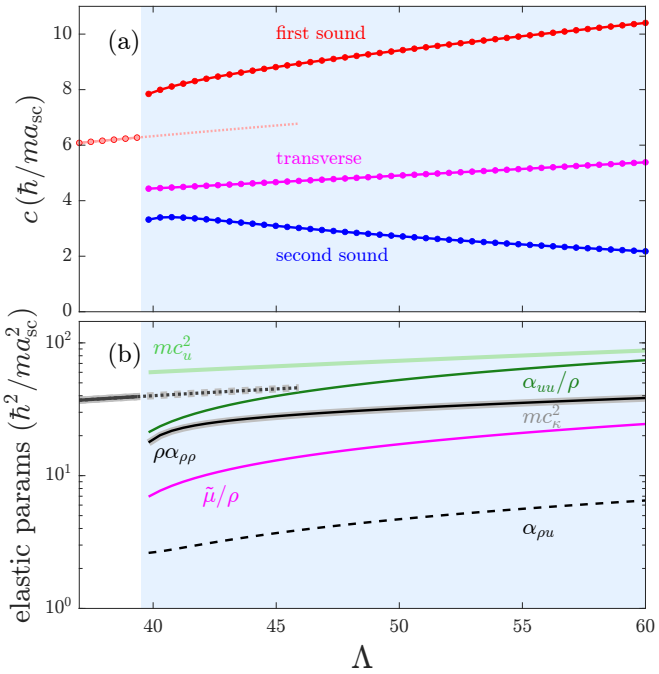

FIG. 5. Elastic properties and speeds of sound for a 2D soft-core BEC at the first-order transition to a supersolid. (a) speeds of sound and (b) elastic parameters. The color code and the quantities are the same as in Fig. 4. The uniform results are continued below the transition (where it is metastable) until it becomes dynamically unstable when the roton softens (at $\Lambda = 46.30$).

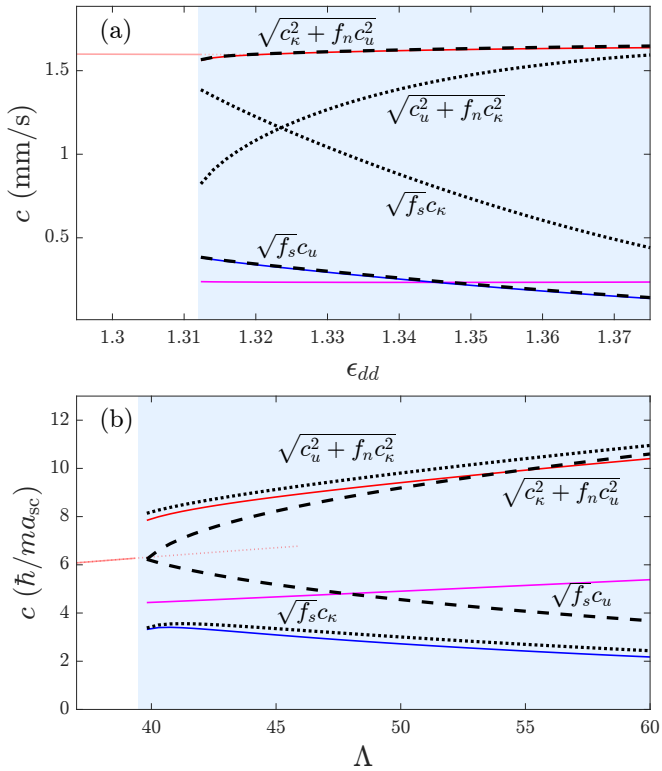

FIG. 6. Comparison of first and second speeds of sound to limiting results for the (a) dipolar and (b) soft-core results. Colored solid and dotted lines are the BdG results for the speeds of sound with the same colors as used in Figs. 4(a) and 5(a). Black dotted and dashed lines indicate the rigid lattice and bulk incompressibility limiting results for c_1 and c_2.

than in the soft-core system. The second sound speed c_2 is always much lower than first sound. Except close to the transition in the soft-core system, c_2 generally decreases as we go deeper into the crystalline phase. This reduction in c_2 is a sign of the reduced superfluidity. Indeed, sufficiently far into the crystalline phase we expect a transition to an isolated droplet crystal and the superfluidity will vanish, although this is beyond the theories we use here (see [46]).

The elastic parameters extracted from the ground state solutions are presented in Figs. 4(b) and 5(b), with f_s appearing in Figs. 1(a) and 2(a). From these parameters we can evaluate the hydrodynamic predictions for the speeds of sound from Eqs. (11)-(13). These results are plotted as symbols in Figs. 4(a) and 5(a) and reveal excellent agreement with the BdG results. It is interesting to consider the role of the density-strain term $\alpha_{\rho u}$, as our results show this term is relatively small, and it is neglected in some treatments [12, 15]. For the dipolar results, dropping this term shifts the hydrodynamic prediction for c_2 down by about 0.3% and the c_1 prediction up by 0.4%. Including the density strain term gives results in better agreement with the BdG results, but both sets of results are almost indiscernible from the BdG results on the scales of our figures.

D. Limiting behavior

To gain a deeper understanding of the behavior of these systems it is useful to introduce the characteristic speeds c_κ and c_u [47], defined as

$$mc_\kappa^2 \equiv \rho \frac{\alpha_{\rho\rho}\alpha_{uu} - \alpha_{\rho u}^2}{\alpha_{uu}} = \frac{1}{\rho\kappa}, \tag{16}$$

$$mc_u^2 \equiv \frac{\alpha_{uu}}{\rho_n}. \tag{17}$$

The first speed is associated with the system compressibility κ [14, 47] and the second is associated with the lattice elastic properties. In the uniform superfluid, with a single gapless excitation branch that dominates the low q density response of the system, c_κ describes the first sound[5]. In the supersolid phase neither of these characteristic speeds correspond to any of the speeds of sound, but they are useful for defining two regimes of behavior. For reference, we indicate these quantities compared to the other elastic parameters in Figs. 4(b) and 5(b). We observe that $c_\kappa \gg c_u$ in the dipolar supersolid,

[5] Here α_{uu} and $\alpha_{\rho u}$ are both zero and Eq. (11) reduces to c_κ.